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Implementation of Naive Bayes classification for Bank Loan prediction

Code :

```
# Naive Bayes - Bank Loan Prediction
```

```
data = [
```

```
    ([1, 1], "Yes"),
```

```
    ([1, 0], "Yes"),
```

```
    ([0, 1], "No"),
```

```
    ([0, 0], "No"),
```

```
    ([1, 1], "Yes"),
```

```
    ([0, 0], "No")
```

```
]
```

```
# 1 = Good income/credit, 0 = Poor income/credit
```

```
test = [1, 1]
```

```
classes = ["Yes", "No"]
```

```
prob = {}
```

```
for c in classes:
```

```
    samples = [x for x, y in data if y == c]
```

```
    p = len(samples) / len(data)
```

```
    for i in range(2):
```

```
        count = sum(x[i] == test[i] for x in samples)
```

```
        p *= (count + 1) / (len(samples) + 2)
```

```
    prob[c] = p
```

```
print("Test Data:", test)
```

```
print("Probability:", prob)
```

```
print("Loan Prediction:", max(prob, key=prob.get))
```

Output :

```
Test Data: [1, 1]
Probability: {'Yes': 0.24, 'No': 0.04000000000000001}
Loan Prediction: Yes

=== Code Execution Successful ===
```